Weekly report

Natalia Trigo Tomasevich

October 30, 2026

From last week

- Complete proof of the fluid model result

From last week

- Complete proof of the fluid model result
- Exploration of the competition model

From last week

- Complete proof of the fluid model result
- Exploration of the competition model
- Insights about the parameters of the problem

Fluid approximation

Recall the state dynamics

$$\dot{x}(t) = D(x(t))\lambda(p_{i(t)} - x(t)) \tag{1}$$

and the provider's problem

$$V(x) = \sup \int_0^\infty e^{-\delta t} D(x(t))(R - c_{i(t)})\, dt, \qquad x(0) = x. \tag{2}$$

Fluid approximation

Recall the state dynamics

$$\dot{x}(t) = D(x(t))\lambda(p_{i(t)} - x(t)) \tag{1}$$

and the provider's problem

$$V(x) = \sup \int_0^\infty e^{-\delta t} D(x(t))(R - c_{i(t)})\, dt, \qquad x(0) = x. \tag{2}$$

The problem with this equation is that the instantaneous payoff are dependent on the product choice. Can we transform the problem so that the two products have the same payoff and differ only on how they move the state?

Transforming the problem

We start from the HJB equation for the fluid model

$$\delta V(x) = D(x) \max_{i \in \{1,2\}} \{R - c_i + \lambda(p_i - x)V'(x)\}. \tag{3}$$

Comparing product 2 vs product 1 we get

$$i = 2 \quad \Longleftrightarrow \quad V'(x) > \frac{\Delta c}{\lambda \Delta p}$$

Transforming the problem

Product 2 has an additional instrantaneous cost

$$D(x)\Delta c$$

But it also increases the drift of the state by

$$\dot{x}_2 - \dot{x}_1 = \lambda(p_2 - p_1) = \lambda\Delta p$$

Therefore,

$$\frac{\text{extra cost}}{\text{extra drift}} = \frac{D(x)\Delta c}{\lambda D(x)\Delta p} = \frac{\Delta c}{\lambda\Delta p}$$

This ratio does not depend on the state. The additional cost per additional unit of state drift is constant. So we can assign a value of $\frac{\Delta c}{\lambda\Delta p}$ to each unif of state accumulation. Once we do that, the cost difference between the products disappears.

Transforming the problem

Adding $\frac{\Delta c}{\lambda \Delta p}$ to the flow payoff and using the dynamics $\dot{x} = D(x)\lambda(p_i - x)$ we get

$$D(x)(R - c_i) + \frac{\Delta c}{\lambda \Delta p}\dot{x} = D(x)\left[R - c_i + \frac{\Delta c}{\Delta p}(p_i - x)\right]$$
$$= D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right]$$

So we have

$$\Rightarrow D(x)(R - c_i) = D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right] - \frac{\Delta c}{\lambda \Delta p}\dot{x}$$

The first term does not depend ont the product choice.

Transforming the problem

Replacing this in the fluid model equation we get

$$V(x) = \sup_{i(\cdot)} \int_0^\infty e^{-\delta t} \left[D(x(t)) \left(R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x(t)) \right) - \frac{\Delta c}{\lambda \Delta p} \dot{x}(t) \right] dt$$

Integrating by parts and doing more math we get

$$V(x) = \frac{\Delta c}{\lambda \Delta p} x + \sup_{i(\cdot)} \int_0^\infty e^{-\delta t} F(x(t)) dt \tag{4}$$

Where

$$F(x) = D(x) \left(R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x) \right) - \frac{\delta \Delta c}{\lambda \Delta p} x$$

Why is this transformation useful?

We can define

$$W(x) = V(x) - \frac{\Delta c}{\lambda \Delta p} x \quad \Rightarrow \quad V'(x) = W'(x) + \frac{\Delta c}{\lambda \Delta p}$$

Why is this transformation useful?

We can define

$$W(x) = V(x) - \frac{\Delta c}{\lambda \Delta p} x \quad \Rightarrow \quad V'(x) = W'(x) + \frac{\Delta c}{\lambda \Delta p}$$

So our condition becomes

$$V'(x) > \frac{\Delta c}{\lambda \Delta p} \quad \Longleftrightarrow \quad W'(x) > 0$$

Why is this transformation useful?

We can define

$$W(x) = V(x) - \frac{\Delta c}{\lambda \Delta p} x \quad \Rightarrow \quad V'(x) = W'(x) + \frac{\Delta c}{\lambda \Delta p}$$

So our condition becomes

$$V'(x) > \frac{\Delta c}{\lambda \Delta p} \quad \Longleftrightarrow \quad W'(x) > 0$$

and the transformed HJB is

$$\delta W(x) = F(x) + \max_i \{\lambda D(x)(p_i - x)\} \tag{5}$$

Why is this transformation useful?

We can define

$$W(x) = V(x) - \frac{\Delta c}{\lambda \Delta p} x \quad \Rightarrow \quad V'(x) = W'(x) + \frac{\Delta c}{\lambda \Delta p}$$

So our condition becomes

$$V'(x) > \frac{\Delta c}{\lambda \Delta p} \quad \Longleftrightarrow \quad W'(x) > 0$$

and the transformed HJB is

$$\delta W(x) = F(x) + \max_i \{\lambda D(x)(p_i - x)\} \tag{5}$$

Now both products face the same payoff $F(x)$ and they differ only in how they move the state.

What we want to prove

Proposition

The fluid problem admits an optimal stationary policy with numbers $0 \leq x_L \leq x_H \leq 1$ such that, away from threshold states,

$$i(x) = \begin{cases} 1, & x < x_L, \\ 2, & x_L < x < x_H, \\ 1, & x > x_H. \end{cases}$$

What we want to prove

Proposition

The fluid problem admits an optimal stationary policy with numbers $0 \leq x_L \leq x_H \leq 1$ such that, away from threshold states,

$$i(x) = \begin{cases} 1, & x < x_L, \\ 2, & x_L < x < x_H, \\ 1, & x > x_H. \end{cases}$$

This is equivalent to proving that $\{x \in [0,1] : W'(x) > 0\}$ is an interval or that we can only see the patterns $1 \to 2$, $2 \to 1$, $1 \to 2 \to 1$ or constant policies but NOT $1 \to 2 \to 1 \to 2$.

How do we prove it

We first study the shape of F and then we use the HJB to restrict the shape of W. Define

$$g(x) = \delta W(x) - F(x) = \max_i \{b_i(x) W'(x)\}$$

where

$$b_i(x) = \lambda D(x)(p_i - x) \qquad \text{(Drifts)}$$

How do we prove it

We first study the shape of F and then we use the HJB to restrict the shape of W. Define

$$g(x) = \delta W(x) - F(x) = \max_i \{b_i(x) W'(x)\}$$

where

$$b_i(x) = \lambda D(x)(p_i - x) \qquad \text{(Drifts)}$$

On a region were product i is being used we have

$$g(x) = b_i(x) W'(x) \quad \Rightarrow \quad W'(x) = \frac{g(x)}{b_i(x)}$$

How do we prove it

We first study the shape of F and then we use the HJB to restrict the shape of W. Define

$$g(x) = \delta W(x) - F(x) = \max_i \{b_i(x) W'(x)\}$$

where

$$b_i(x) = \lambda D(x)(p_i - x) \qquad \text{(Drifts)}$$

On a region were product i is being used we have

$$g(x) = b_i(x) W'(x) \quad \Rightarrow \quad W'(x) = \frac{g(x)}{b_i(x)}$$

differentiating this with respect to x we get

$$g'(x) = \delta W'(x) - F'(x) = \frac{\delta}{b_i(x)} g(x) - F'(x)$$

Connection between W and F

Consider x^* away from p_1 and p_2 such that $W'(x^*) = 0$. Then,

$$g(x^*) = b_i(x^*)W'(x^*) = 0$$

Connection between W and F

Consider x^* away from p_1 and p_2 such that $W'(x^*) = 0$. Then,

$$g(x^*) = b_i(x^*)W'(x^*) = 0$$

since $b_i(x^*) \neq 0$,

$$g'(x^*) = \frac{\delta}{b_i(x^*)}g(x^*) - F'(x^*) = -F'(x^*)$$

So at a possible switching point, the sign of F' tells us the direction in which g can cross zero.

Lemma

Lemma

The set $\{x \in [0,1] : F'(x) > 0\}$ is an interval, possibly empty or meeting a boundary. If nonempty, F decreases before this interval, increases within it, and decreases after it, with absent segments omitted. If it is empty, product 1 is optimal everywhere.

This proof is in the progress notes.

What we want to prove now

Proposition

Suppose $0 < p_1 < p_2 < 1$, and $\delta, \lambda > 0$. Allow relaxed controls, so that any state in $[p_1, p_2]$ can be held fixed. Suppose that, for some $0 \leq a < m \leq 1$,

$$F'(x) < 0 \quad \text{on } (0, a) \cup (m, 1), \qquad F'(x) > 0 \quad \text{on } (a, m), \tag{6}$$

with any interior zeros of F' being simple. Then an optimal policy can be selected which uses product 2 on at most one interval and product 1 outside it. Thus, as the state increases, the only nonconstant action patterns are

$$1 \to 2, \qquad 2 \to 1, \qquad 1 \to 2 \to 1. \tag{7}$$

Proof

First we notice that the sign of the drifts $b_i(x)$ is determined by the position of x:

$$b_1(x) > 0\,, \quad b_2(x) > 0 \quad \text{for } x < p_1 \tag{8}$$
$$b_1(x) < 0\,, \quad b_2(x) > 0 \quad \text{for } p_1 < x < p_2 \tag{9}$$
$$b_1(x) < 0\,, \quad b_2(x) < 0 \quad \text{for } x > p_2 \tag{10}$$

Since $g = b_i W'$ for any branch of product i, we have that the sign of g is related to the sign of W', specifically:

$$\text{sign}(g) = \text{sign}(W')\,, \quad \text{for } x < p_1 \tag{11}$$
$$\text{sign}(g) = -\text{sign}(W')\,, \quad \text{for } x > p_2 \tag{12}$$

Outside $[p_1, p_2]$

At an interior zero of g with $F' \neq 0$, we have that $g' = -F'$, so

	$F' < 0$	$F' > 0$
$x < p_1$	$1 \to 2$	$2 \to 1$
$x > p_2$	$2 \to 1$	$1 \to 2.$

(*)

The case where $F' = 0$ is not a problem because F' has simple zeros, when this happens and $g = 0$, it does without changing sign.

Inside $[p_1, p_2]$

- Consider the case $m > p_1$ and define $h = min\{m, p_2\}$

Inside $[p_1, p_2]$

- Consider the case $m > p_1$ and define $h = min\{m, p_2\}$
- In this region we have that

$$b_1(x) < 0 < b_2(x)$$

Inside $[p_1, p_2]$

- Consider the case $m > p_1$ and define $h = min\{m, p_2\}$
- In this region we have that

$$b_1(x) < 0 < b_2(x)$$

- Product 1 moves the state towards p_1 and product 2 moves it towards p_2

Inside $[p_1, p_2]$

- Consider the case $m > p_1$ and define $h = min\{m, p_2\}$
- In this region we have that

$$b_1(x) < 0 < b_2(x)$$

- Product 1 moves the state towards p_1 and product 2 moves it towards p_2
- We define $L(x)$ as the value of moving towards p_1 and $H(x)$ as the value of moving towards h. They satisfy the Bellman equations:

$$\delta L = F + b_1 L' \qquad \delta H = F + b_2 H'$$

Inside $[p_1, p_2]$

- Consider the case $m > p_1$ and define $h = min\{m, p_2\}$
- In this region we have that

$$b_1(x) < 0 < b_2(x)$$

- Product 1 moves the state towards p_1 and product 2 moves it towards p_2
- We define $L(x)$ as the value of moving towards p_1 and $H(x)$ as the value of moving towards h. They satisfy the Bellman equations:

$$\delta L = F + b_1 L' \qquad \delta H = F + b_2 H'$$

- On the decreasing part of F,moving left toward p_1 raises the future flow payoff. On the increasing part, moving right toward p_2 does the same.

Inside $[p_1, p_2]$

- That gives us

$$\delta \max\{L(x), H(x)\} > F(x)$$

Inside $[p_1, p_2]$

- That gives us
$$\delta \max\{L(x), H(x)\} > F(x)$$
- If L wins because of $b_1(x) < 0 < b_2(x)$ we have that $L' < 0$

Inside $[p_1, p_2]$

- That gives us
$$\delta \max\{L(x), H(x)\} > F(x)$$
- If L wins because of $b_1(x) < 0 < b_2(x)$ we have that $L' < 0$
- If H wins because of $b_1(x) < 0 < b_2(x)$ we have that $H' > 0$

Inside $[p_1, p_2]$

- That gives us
$$\delta \max\{L(x), H(x)\} > F(x)$$
- If L wins because of $b_1(x) < 0 < b_2(x)$ we have that $L' < 0$
- If H wins because of $b_1(x) < 0 < b_2(x)$ we have that $H' > 0$
- Then, considering an interior crossing $x^* \in (p_1, h)$ such that $L(x^*) = H(x^*)$, we have that
$$\delta L(x^*) - F(x^*) = \delta H(x^*) - F(x^*) > 0$$
and with $b_1(x^*) < 0 < b_2(x^*)$ we have
$$L'(x^*) = \frac{\delta L(x^*) - F(x^*)}{b_1(x^*)} < 0, \qquad H'(x^*) = \frac{\delta H(x^*) - F(x^*)}{b_2(x^*)} > 0$$

Inside $[p_1, p_2]$

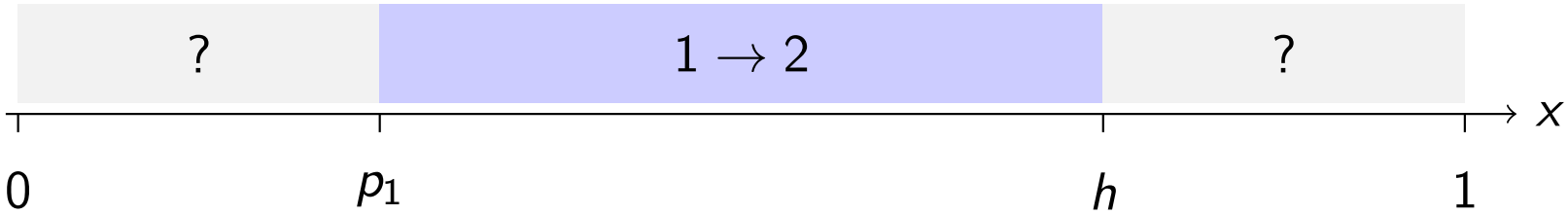
- That gives us
$$\delta \max\{L(x), H(x)\} > F(x)$$
- If L wins because of $b_1(x) < 0 < b_2(x)$ we have that $L' < 0$
- If H wins because of $b_1(x) < 0 < b_2(x)$ we have that $H' > 0$
- Then, considering an interior crossing $x^* \in (p_1, h)$ such that $L(x^*) = H(x^*)$, we have that
$$\delta L(x^*) - F(x^*) = \delta H(x^*) - F(x^*) > 0$$
and with $b_1(x^*) < 0 < b_2(x^*)$ we have
$$L'(x^*) = \frac{\delta L(x^*) - F(x^*)}{b_1(x^*)} < 0, \qquad H'(x^*) = \frac{\delta H(x^*) - F(x^*)}{b_2(x^*)} > 0$$
- Then $(H - L)'(x^*) > 0$, which means that H wins immediately to the right of x^* and L wins immediately to the left.

Inside $[p_1, p_2]$

- The second part of this case is to prove that the maximum of L and H is the optimal policy, that I won't show here but you can find in the progress notes.
- With this we have that if there is any crossing it must be unique and going from L to H.

Inside $[p_1, p_2]$

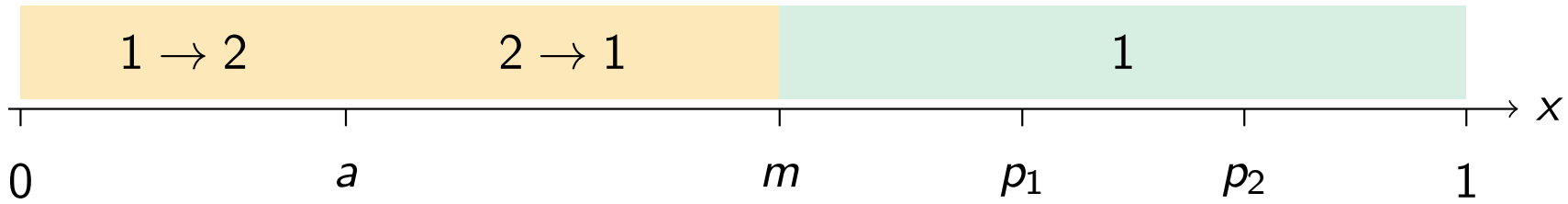
- The second part of this case is to prove that the maximum of L and H is the optimal policy, that I won't show here but you can find in the progress notes.
- With this we have that if there is any crossing it must be unique and going from L to H.



When $m \leq p_1$

- From any state $x \geq 0$ no product can move the state below m and since F is decreasing above m, product 1 generates the smallest trajectory so it is optimal
- Below m we have $x < p_1$ so from the crossing table we have at most $1 \to 2$ when F is decreasing and $2 \to 1$ when F is increasing.

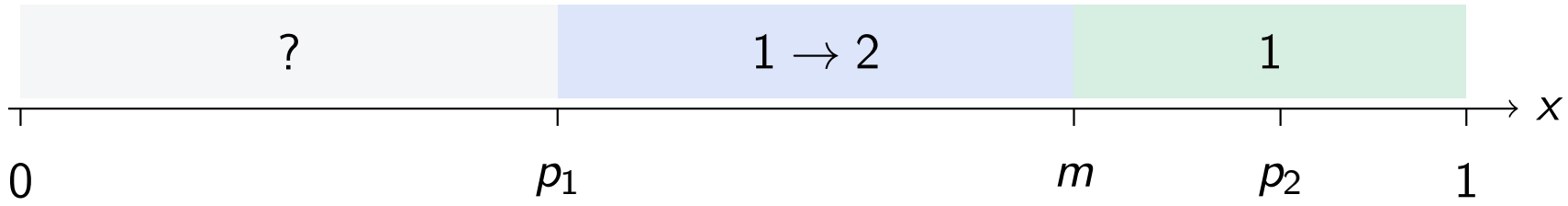
Then we have



When $p_1 < m \leq p_2$

- For $x > m$ we want to go back to m and since $b_1(x) < b_2(x)$ at every state, the trajectory generated by poduct 1 is smaller than any other trajectory. Hence, product 1 is optimal
- On $[p_1, m]$ we already saw that the only crossing that can exists is $1 \to 2$ (case h=m)

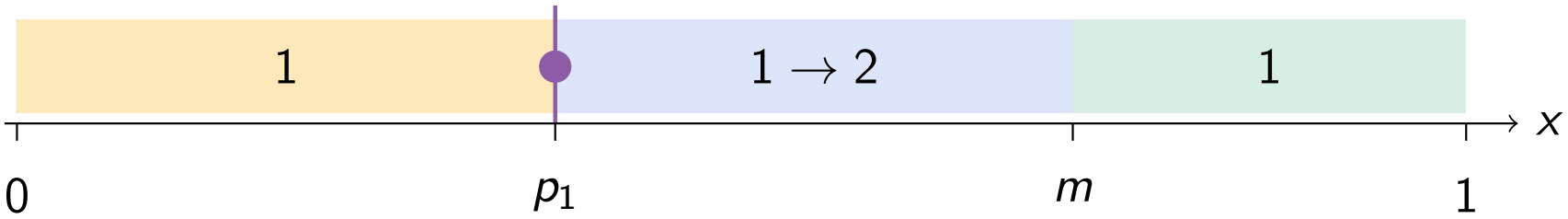
With these results we have



When $p_1 < m \leq p_2$

When $a > p_1$ and L wins at p_1

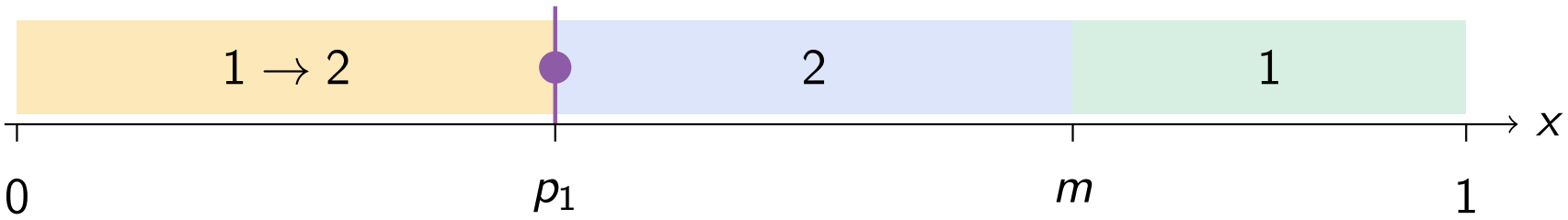
- Below p_1 we have $F' < 0$ so the only possible crossing is $1 \to 2$
- Here $W(p_1) = L(p_1) = F(p_1)/\delta \to g(p_1) = 0$
- A contradiction argument shows that product 1 must be optimal immediately to the left of p_1
- A product 2 region below p_1 would require a $2 \to 1$ crossing and that is not allowed



When $p_1 < m \leq p_2$

When $a > p_1$ and H wins at p_1

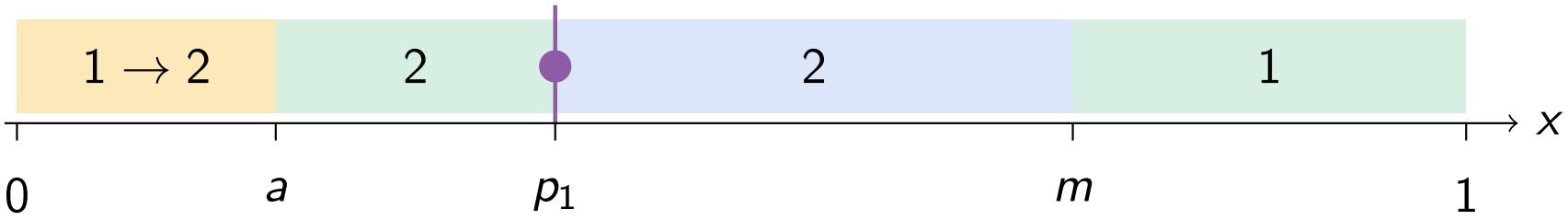
- Here we have $W(p_1) = H(p_1) > L(p_1) = F(p_1)/\delta$
- Then $g(p_1) > 0$ and by continuity, $g(x) > 0$ to the left of p_1
- Since $0 < b_1(x) < b_2(x)$, $W' > 0$, hence product 2 is optimal to the left of p_1
- Again, the table allows crossing from $1 \to 2$ only so any product 2 region on the left can begin at most once and must continue up to p_1



When $p_1 < m \leq p_2$

When $a < p_1$

- On (a, p_1), we have $F' > 0$ so moving toward p_1 under product 1 raises the future payoff, so $L > F/\delta$ and hence $g > 0$
- Since $0 < b_1(x) < b_2(x)$ we have $W' > 0$ and hence product 2 is optimal.
- On $(0, a)$, $F' < 0$ so only $1 \to 2$ crossing is allowed
- Therefore any product 2 region below a must attach to the one on (a, p_1)



When $p_1 < m \leq p_2$

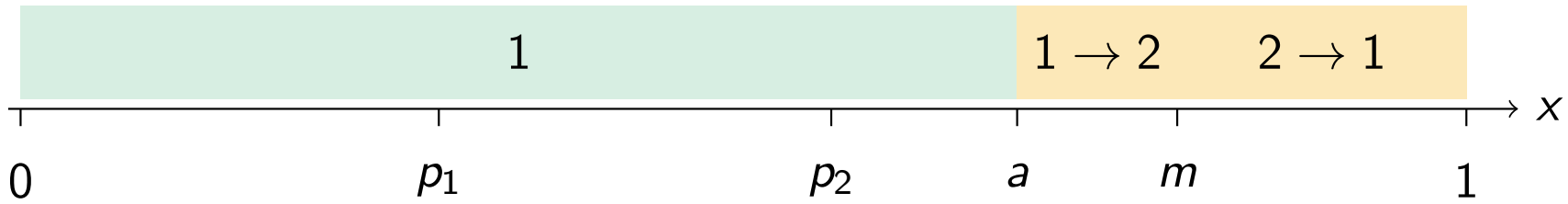
When $a = p_1$

- a is the valley of F
- Moving toward the higher destination yields a larger future payoff
- Hence, the high branch wins at p_1 and by continuity, product 2 is optimal to the left of p_1
- Since $F' < 0$ below p_1 the table shows the only possible crossing is $1 \to 2$



When $m > p_2$

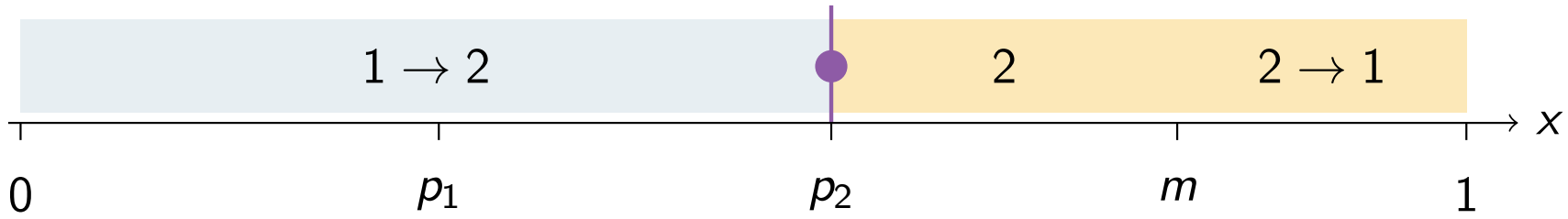
- In $[p_1, p_2]$ the destination comparison is straightforward
- Below p_1 we also saw it already
- If $a \geq p_2$, any trajectory starting at $x < a$ remains below a and product 1 is optimal
- For $x > a$ we have the table that allows only $1 \to 2$ crossing on (a, m) and only $2 \to 1$ crossing on $(m, 1)$ hence product 2 is optimal at most on at most one interval



When $m > p_2$

When $a < p_2$

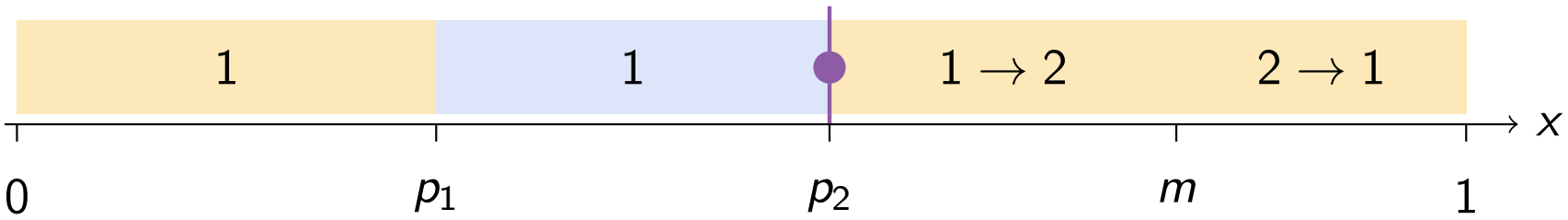
- We have $F'(x) > 0$ on (p_2, m), so the table allows crossing $1 \to 2$
- If H wins at p_2 by a contradiction argument we get that product 2 is optimal to the right of p_2



When $m > p_2$

When $a < p_2$

- Since all crossings inside $[p_1, p_2]$ have orientation $L \to H$, if L wins at p_2 then L wins throughout the interval
- On (p_2, m) we have $F'(x) > 0$, so the table allows crossing $1 \to 2$
- On $(m, 1)$ we have $F'(x) < 0$, so the table allows crossing $2 \to 1$
- Then product 2 can be optimal on at most one interrval



Endpoint ties

The crossing argument used before relies on

$$W'(x) = \frac{g(x)}{b_i(x)}$$

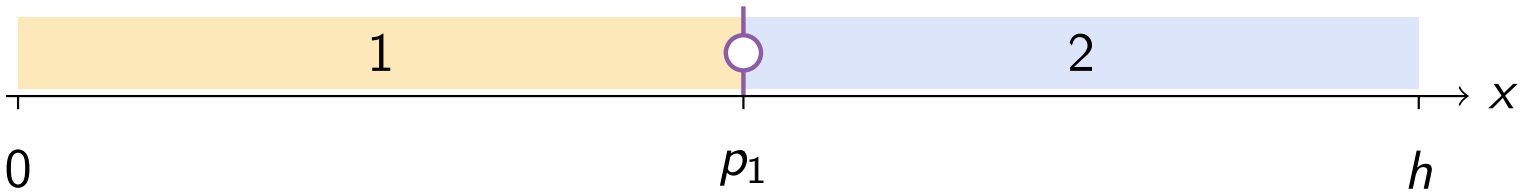
But at the endpoints we have

$$b_1(p_1) = 0 \quad , \quad b_2(p_2) = 0$$

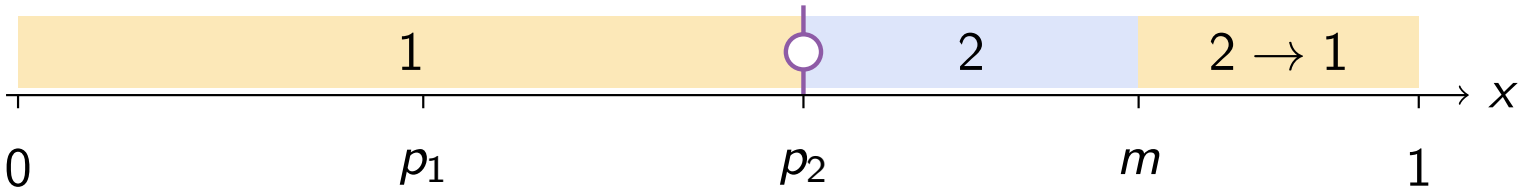
so this argument cannot be applied directly. The goal then is to determine which product is optimal immediately on each side of the tie

Endpoint ties

- At a tie at p_1, the policy switches from product 1 to product 2.



- At a tie at p_2, the policy also switches from product 1 to product 2.



With all this

- All these arguments show that the only patterns that can occur are $1 \to 2$, $2 \to 1$, $1 \to 2 \to 1$ and constant policies. Hence, the optimal policy uses product 2 at most one interval
- This proposition + the proof of existence of thresholds is the characterization that we can provide for the optimal policy